



Evaluation of immunomodulatory (humoral as well as cell-mediated) and cytokines (TNF- α & IL-10) regulating potential of *Neolamarckia cadamba* fruit extract in Wistar albino rats

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Abstract

Current work has been designed to investigate the immunomodulatory efficacy with particular reference to humoral, cell-mediated and cytokine-modulating potential of hot aqueous extract of *Neolamarckia cadamba* (HAENC) fruits in Wistar albino rats. The effect of different concentrations of HAENC fruits over cell-mediated immune response was assessed using six groups (Gp-I as control, Gp-II with 20 $\mu\text{g/mL}$, Gp-III with 50 $\mu\text{g/mL}$, Gp-IV with 100 $\mu\text{g/mL}$, Gp-V with 250 $\mu\text{g/mL}$, and Gp-VI with 500 $\mu\text{g/mL}$) of Wistar albino rats having six animals in each. The amount of tumor necrosis factor- α (TNF- α) and interleukin (IL)-10 was measured by sandwich ELISA with different concentrations of HAENC (50–500 $\mu\text{g/mL}$) in splenocyte culture supernatant and their expression was determined by qRT-PCR. Humoral immune response was determined by measuring the serum antibody titer of Wistar albino rats against *Salmonella typhimurium* ‘O’ antigen using four groups containing six animals each (Gp-I as control, Gp II, III & IV were respectively fed orally with 125, 250, and 500 mg/Kg body weight using HAENC fruits). LC–MS analysis suggested the presence of cadambine, chlorogenic acid, cadambagenic acid, stearic acid, octadecanoic acid ethyl ether, and 7-hydroxy-5,2'-4'-trimethoxyflavonon in the extract based on their m/z ratio. The result suggested significant ($p < 0.01$) dose-dependent proliferation of Concanavalin A (Con A)-treated splenocytes, depicting cell-mediated immunostimulatory potential of HAENC fruits. A dose-dependent significant decrease ($p < 0.01$) was found in the amount of TNF- α and IL-10 was found to increase significantly ($p < 0.01$) as extract concentrations increased. TNF- α and IL-10 expressions were confirmed at the molecular level by qRT-PCR analysis of mRNA transcripts of TNF- α and IL-10 genes. Fold expression of TNF- α and IL-10 gene was 0.197 and 3.58 at 250 $\mu\text{g/mL}$, 0.02 and 20.11 at 500 $\mu\text{g/mL}$ concentrations of HAENC respectively in comparison to control. Serum antibody titer was significantly increased ($p < 0.01$) in animals fed with different doses of HAENC fruits. The present study suggested the anti-inflammatory effect of HAENC fruits which also influences the networking of cytokines, implying that it may play a role in regulating the activity of the host's immune system and can serve as a potent herbal drug with immuno-stimulatory potential.

Keywords *Neolamarckia cadamba* · Cell-mediated immune response · Humoral immune response · Tumor necrosis factor- α · IL-10 · Anti-inflammatory

Introduction

There has always been a connection between life, sickness, and plants since the dawn of time. Diseases and cures were studied by primitive men. There is no evidence that

individuals in ancient times were interested in synthetic medicines for their ailments, but they did try to make do with what they could easily obtain. The most basic clothing they could locate was among the saplings and animals in the area. They started with plants and discovered that the majority of them could be eaten, while others were poisonous or medicinally valuable (Lyons and Petrucelli 1987).

Since the beginning of human civilization, herbal formulations have secured their appropriate position in traditional medicines as paraphrased by Greek, Chinese, Tibetan, Indian, and Mediterranean Ayurveda and other ancient medical systems. Now these medicinal herbs are getting the

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focus of researchers and moving from fringe to mainstream and placed under the rationalist with a greater number of people seeking health approaches due to lesser side effects caused by synthetic medicines (Pengelly 2020).

The fundamentally innovative method for the cure of infectious diseases is to modulate host immune responses to boost contagious agent clearance and reduce tissue damage due to inflammation. A class of herbal medicine called immunomodulators changes the activity of immune function by dynamically regulating informational molecules such as cytokines. Immunomodulators are medicines that are used to alter immune responses (Easton et al. 2009). Immunomodulators are divided into three categories in clinical practice: immunostimulants, immunoadjuvants, and immunosuppressants. Immunoadjuvants are used to improve vaccine effectiveness (Agarwal and Singh 1999). Immunostimulants are general because they increase the body's resistance to infection. They can function both through innate and adaptive immune responses. Immunosuppressants are used to treat autoimmune disorders/diseases as well as rejection of various types of organ transplantation. Clinical investigations have suggested that phytotherapy could be effective in treating a variety of disorders caused by immune system malfunction by regulating informational molecules such as hormones, cytokines, chemokines, neurotransmitters, and other peptides dynamically. They can also be utilized as part of a chemotherapy treatment plan. The adoption of molecular tools to investigate the networking of biomolecules (cytokines) expressed under various circumstances prevailing in the host from time to time has revolutionized immunology research (Owen et al. 2013).

Presently, more than 200 different cytokines have been identified, and their potential applications offer the possibility of targeted therapies to modify immune responses in organ transplantation, infectious diseases, allergic reactions, inflammatory diseases, and cancer treatment (Gulati et al. 2016). Pro-inflammatory cytokines will cause immune cells to become activated, produce more cytokines, and release them. Although inflammatory cytokines have been linked to several inflammatory disorders, the details of these cytokines' precise functions have not always been reported in the literature. In addition to being therapeutically useful characteristics for treatments, they can be employed as biomarkers to identify or track the progression of disease. However, it is not always evident what exactly their role is (Kany et al. 2019).

Medicinal plants are abundant on the Indian subcontinent and are employed in conventional medical practices. Approximately 20,000 medicinal plants have been identified in India; however, only 7000–7500 of these are used by traditional healers (Saini et al. 2022). Previously, a number of medicinal plants such as *Silybum marianum*, *Matricaria chamomilla*, *Calendula officinalis*

and *Cichorium intybus* had been studied and exhibited immunomodulatory activity. Among all medicinal plants *Neolamarckia cadamba* also secured its potential medicinal value due to the presence of several phytochemicals such as flavonoids, alkaloids, coumarins, terpenoids, diterpenoids, triterpenes, glycosides, sterols, amides, and fatty acids making it accountable for various pharmacological activities such as antimicrobial, antipyretic, anti-inflammatory, analgesic, anticancerous, antidiabetic, hypolipidemic, antihepatotoxic, antidiarrhoeal, diuretic, laxative, antioxidant, immunomodulatory, and wound healing (Pandey and Negi 2016). This evergreen subtropical tree belonging to the family Rubiaceae recognized with different synonyms as *Neolamarckia cadamba*, *Anthocephalus cadamba* (Roxb.) Miq., *Anthocephalus chinensis* (Lamk.), *Anthocephalus macrophyllus* Havil., *Nauclea megaphylla*, *Nauclea cadamba* (Roxb.), *Sarcocephalus cadamba* Kurz, *Samama cadamba* Kuntze etc. Biomolecules must be studied as markers that are more clearly linked to health and disease. This study was carried out by WHO criteria on herbal medicines to provide scientific proof to historical beliefs that *Neolamarckia cadamba*, a sacred plant, has vast therapeutic potential in treating numerous ailments. It is a historical tree of the Rubiaceae family that has been used in folklore medicine to cure fever, uterine problems, anemia, blood illnesses, skin diseases, eye inflammation, diarrhea, leprosy, dysentery, and stomatitis. Talking about *N. cadamba*, Khandelwal et al. (2019) reported cytokine modulating potential of hot aqueous extract of *N. cadamba* leaves using the Wistar albino rat model (Khandelwal et al. 2019).

This plant contains the number of phytochemicals and secondary metabolites (viz., cadambagenic acid, cadamine, quinovic acid, β -sitosterol, cadambine, etc.) responsible for its various biological and pharmacological activities (Dwivedi et al. 2015) such as anti-hepatotoxic activities (Kapil et al. 1995), antioxidant (Umachigi et al. 2007; Kumar et al. 2010; Alam et al. 2011), anti-inflammatory (Bachhav et al. 2009; Chandrashekar and Prasanna 2009; Mondal et al. 2011), antidiabetic (Gurjar et al. 2010; Chandrashekar and Prasanna 2009; Sanadhya et al. 2013), anti-helminthic (Acharyya et al. 2011) and antimicrobial (Chandrashekar and Prasanna 2009; Sanadhya and Durve 2014; Khandelwal et al. 2016).

Till now no study has been proposed to investigate humoral, cell-mediated immune potential as well as cytokines regulating efficacy of *N. macadamia* fruit. Therefore, the present study has been carried out to find the Immunomodulatory (humoral and cell-mediated) and cytokines modulating potential of *Neolamarckia cadamba* fruit at the Laboratory of Department of Biotechnology, GLA University, Mathura, Uttar Pradesh during the year 2021–2022.



Materials and methods

Collection and identification of fruits

The fresh, mature fruits of *Neolamarckia cadamba* were harvested from wild and cultivated populations from Mathura (27°31'2" N–77°39'26" E) and Vrindavan areas (27°33'39" N–77°41'14" E) of Uttar Pradesh during May 2021. The fruits were identified and authenticated by Agharkar Research Institute, Pune with authentication ID AUTH 21–85. Authenticated fruits were collected in mass, cleaned, and oven-dried at 48⁰ C in the cabinet hot air dryer until constant weight. Dried fruits were powdered in a grinder and allowed to shade-dry for two weeks before the extraction process.

Preparation of hot aqueous extract of *N. cadamba* fruit by reflux extraction method

10 g of *N. cadamba* fruit powder was combined with 400 mL of triple distilled water in a 500 mL round bottom flask with a cooling condenser and heated at 95 °C for 2 h. After completion extract was filtered and residue was three times rinsed with extract solvent (Zhang et al. 2018).

The solution present in the collecting flask was evaporated in a rotary evaporator under controlled temperature and reduced pressure further Storage at 4 °C for further use.

LC–MS analysis of compounds presented in aqueous extract of fruits of *N. cadamba*

The extract of *N. Cadamba* fruits was subjected to preliminary screening of phytochemical analysis and the compounds were identified according to their *m/z* ratio where an electrospray ionization (ESI) source was used. The chromatographic separation of metabolites was achieved on water, Acquity ACCUCORE C18 column (150 × 4.6, 2.6µm), and a flow rate of 0.600 mL/min. Four solvents used were: solvent A = % 80.0 H₂O, solvent B = % 20.0 CAN (Ceric ammonium nitrate), solvent C = % 0.0 MEOH (Methanol), and solvent D = % 0.0 5 mM AA. Full-scan data were recorded in ESI-negative and ESI-positive ion modes, from a mass range of *m/z* 150–750 Th.

Determination of cell-mediated immune response of aqueous fruit extract of *N. cadamba* on Wistar albino rat's splenocyte proliferation/inhibition

The in vitro effect of HAENC fruits on splenocyte proliferation/inhibition of rat spleen cells at five concentrations,

i.e., 20, 50, 100, 250, and 500 µg/mL, was investigated as per the method suggested (Goel et al. 2010).

Splenocyte proliferation was done using MTT [3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide] assay. Briefly, the spleen cells, were seeded in a 96-well plate (concentration adjusted to 200 µL of 2×10^6 spleen cells/well) in RPMI-1640 medium with 10% fetal bovine serum (FBS) and 5 µg/mL optimum concentration of mitogen concanavalin A (Con A) used to enhance the immune response (Goel et al. 2010). Finally, the cells were treated with different concentrations (20 µg/mL, 50 µg/mL, 100 µg/mL, 250 µg/mL, and 500 µg/mL) of HAENC fruits. Now the cells were allowed to incubate at 37 °C in a humidified atmosphere at 5% CO₂ in a CO₂ incubator for 72 h. Control cells were only pretreated with an optimum concentration of Con A.

After incubation 20 µL of MTT solution (5 mg/mL) was added into each well which resulted in the formation of formazon crystals due to a reduction in MTT. Now plate was again incubated for 4 h in optimum concentrations. The supernatant was removed and the plate was air-dried. Now crystals were dissolved by the addition of 100 µL of dimethyl sulfoxide (DMSO) in each well and optical density (OD) was taken at dual wavelength 560–670 nm by ELISA reader.

In vitro effect of aqueous fruit extract of *N. cadamba* on cytokine induction

The procedure for cytokine assay was identical to that for splenocyte proliferation assay, with the exception that spleen cell culture was incubated for 48 h before the collection of supernatant for the determination of Th1 cytokines (TNF-α) and Th2 cytokines (IL-10). The quantification of cytokine in cell culture was performed by the kit (R & D system) protocol.

A standard graph of both cytokines was created by kit (R & D system) protocol using various concentrations of TNF- α and IL-10. Using the standard curve for each cytokine, the various cytokines found in the culture supernatant were quantified. OD was measured at dual wavelength (450–570 nm).

Expression of Th1 cytokine (TNF-α) and Th2 cytokine (IL-10) at molecular level by Real Time PCR

The threshold cycle (*C_T*) or Quantification cycle (*C_q*) is the value at which amplification of the target gene gets started after crossing a cut-off value, which will be inversely proportional to the amount of the target gene. mRNA expressions of cytokines were determined using a cDNAdirect kit from GeNei™ which has an optimized procedure to synthesize first-strand cDNA directly from cultured mammalian cells without isolating RNA. The synthesized first strand

of cDNA is ready to be used in PCR, RT-PCR, and other downstream applications such as quantifying mRNA levels from a small number of cells. In this procedure, cultured cells are washed in PBS to remove cell culture medium and extracellular contaminants. Cells are then incubated with a specially designed lysis buffer at 75 °C to rupture the cells and inactivate the endogenous RNases. The crude cell lysate is treated with DNase I to degrade the genomic DNA and is ready to use for first-strand cDNA synthesis and PCR using a cDNA direct kit from GeNei™. PCR conditions were held at 50 °C (for 2 min), activation at 94 °C (for 5 min) followed by 39 cycles of denaturation at 94 °C for 30 s, annealing at 54 °C for 30 s and extension at 72 °C for 30 s. The final extension was done at 72 °C (7 min) after 39 cycles of amplification. Melt curve analysis starts from 65 to 90 °C for 20 min.

The differential expression of TNF- α , IL-10 cytokines, and β -actin (endogenous control) in splenocytes was carried out using qRT-PCR (Quantitative SYBR green real-time PCR) using scientific primers. The sequences of primers were as follows:

IL-10:	5'- GCTCAGCA CTGCTATGT TGC-3'	(Forward primer)
	5'-TTCATGGC CTTGTAGA CACC-3'	(Reverse primer)
TNF- α :	5'-CCA-CGTCGTAGCAAACCA-CCAAG-3'	(Forward primer)
	5'-CAG-GTACAT-GGGCTCATAACC-3'	(Reverse primer)
β -actin:	5'-TGGAGAAGAGCTATGAGC TGC-3'	(Forward primer)
	5'-TCCACACAGAGTACTTGCGC-3'	(Reverse primer)

Expression of the target gene was calculated from the formula

Fold change of target genes (TNF- α & IL-10) = $2^{-\Delta\Delta C_T}$ method (Schmittgen and Livak 2008).

$\Delta\Delta C_T$ = (Target gene-Target internal control) – (Untreated control gene-Untreated internal control). Samples were taken as triplet.

Determination of humoral immune response against *S. Typhimurium* "O" antigen

Salmonella typhimurium 'O' antigen was used to study the humoral immune response in all four groups, including control and those fed with 125/250/500 mg/kg body weight of *N. cadamba* fruit extract for 21 days (Khandelwal et al. 2018).

Immunization schedule

Experimental rats were divided into four groups with six animals each. Group I (control) rats were subcutaneously

inoculated with *S. Typhimurium* "O" antigen but no plant extract. Albino rats in groups II, III, and IV were fed 125 mg/250 mg/500 mg/kg body weight of aqueous fruit extract of *N. cadamba* orally for 21 days and then immunized with 'O' antigen along with Freund's complete and incomplete adjuvants to boost immunization, as per immunization schedule (Khandelwal et al. 2018). Blood serum was collected one week after the last dose of 'O' antigen to determine the antibody titer against *Salmonella* 'O' antigen by indirect ELISA using anti-rat IgG-HRP conjugated antibody.

Determination of antibody titer by Indirect ELISA

The ELISA test was conducted according to standard protocol. Salmonella antigen (100 μ l) was used to coat the wells of a polystyrene microtiter plate (Nunc) overnight at 4 degrees Celsius. After three washes with PBS (pH = 7.2, 0.01 M) containing 0.05 percent tween-20. The blocking was accomplished by incubating 3% bovine serum albumin dissolved in PBS at room temperature for two hours. From 1 to 11 microtiter plate wells, diluted serum samples (1:10 to 1:110,240) were added. Rabbit anti-rat immunoglobulin-G (IgG) conjugated with horseradish peroxidase (HRP) and substrate solution TMB was added serially to each well of an ELISA plate, except the 12th well, which served as a control. Each well-received 50 μ L of 1 M sulphuric acid to prevent further color development. The measured OD range was 450–570 nm. Each step was incubated for two hours at 37 °C, and three washes with washing buffer were performed after each step.

Statistical analysis

To find significant variations between treatment means, an analysis of variance (ANOVA) was performed using SPSS version 20.0 software and DMRT at $p < 0.05$ and 0.01 respectively. The mean $\pm$ SEM is used to express values.

Results

LC–MS analysis of compounds presented in aqueous extract of the fruit of *N. cadamba*

LC–MS analysis of *N. cadamba* fruit extract suggested the presence of certain phytochemical compounds. Results were based on the m/z ratio of the compound observed in the form of a peak on a particular retention time. $[M + H]^+$ adduct with a 546 m/z ratio was found at retention times 6.90, 7.37 which will suggest the presence of Isohydrocadambine or dihydrocadambine in the sample. Chlorogenic acid was found in the form of $[M + H]^-$ at RT 15.56 with a 354 m/z ratio. Stearic acid found at RT 17.34 losing one electron as

in the form of $[M+H]^+$ with 284 m/z ratio. Cadambagenic acid, Octadecanoic acid ethyl ether, and 7-hydroxy-5,2'-4'-trimethoxyflavonon were also found with 486, 312, and 328 m/z ratios, respectively, at particular retention times (see Fig. 1).

The cell-mediated immune response of aqueous fruit extract of *N. cadamba* on Wistar albino rat by splenocyte proliferation/inhibition

Aqueous fruit extract, in a range of concentrations, including 20, 50, 100, 250, and 500 $\mu\text{g/mL}$, was discovered to be safe and non-toxic for splenocytes and was subsequently used for cytokine assay. According to an in vitro study, HAENC fruits caused spleen cell culture to proliferate by 27.78%, 33.25%, 35.73%, 45.64%, and 64.84% respectively. Hot aqueous extract of *N. cadamba* fruit demonstrated dose-dependent splenocyte proliferation, according to the study (Fig. 2).

Effects of aqueous fruit extract on induction of TH1 (TNF- α) and TH2 (IL-10) cytokines

Results showed that there was a dose-dependent decrease in TNF- α with an increase in IL-10 production with increasing concentrations of HAENC fruit. Percent inhibition/stimulation of cytokine production was found to be maximum at 500 $\mu\text{g/mL}$ and minimum at 50 $\mu\text{g/mL}$ extract concentrations (Table 2). Results showed upregulation of IL-10 and downregulation of TNF- α respectively.

Expression of Th1 cytokine (TNF- α) and Th2 cytokine (IL-10) at molecular level by Real Time PCR

Relative fold expression of TNF- α and IL-10 gene was determined by $2^{-\Delta\Delta\text{CT}}$ method. A comparative analysis is given in Fig. 3.

1: MS ES+ :BPI Smooth (SG, 4x2)

7.2e+006

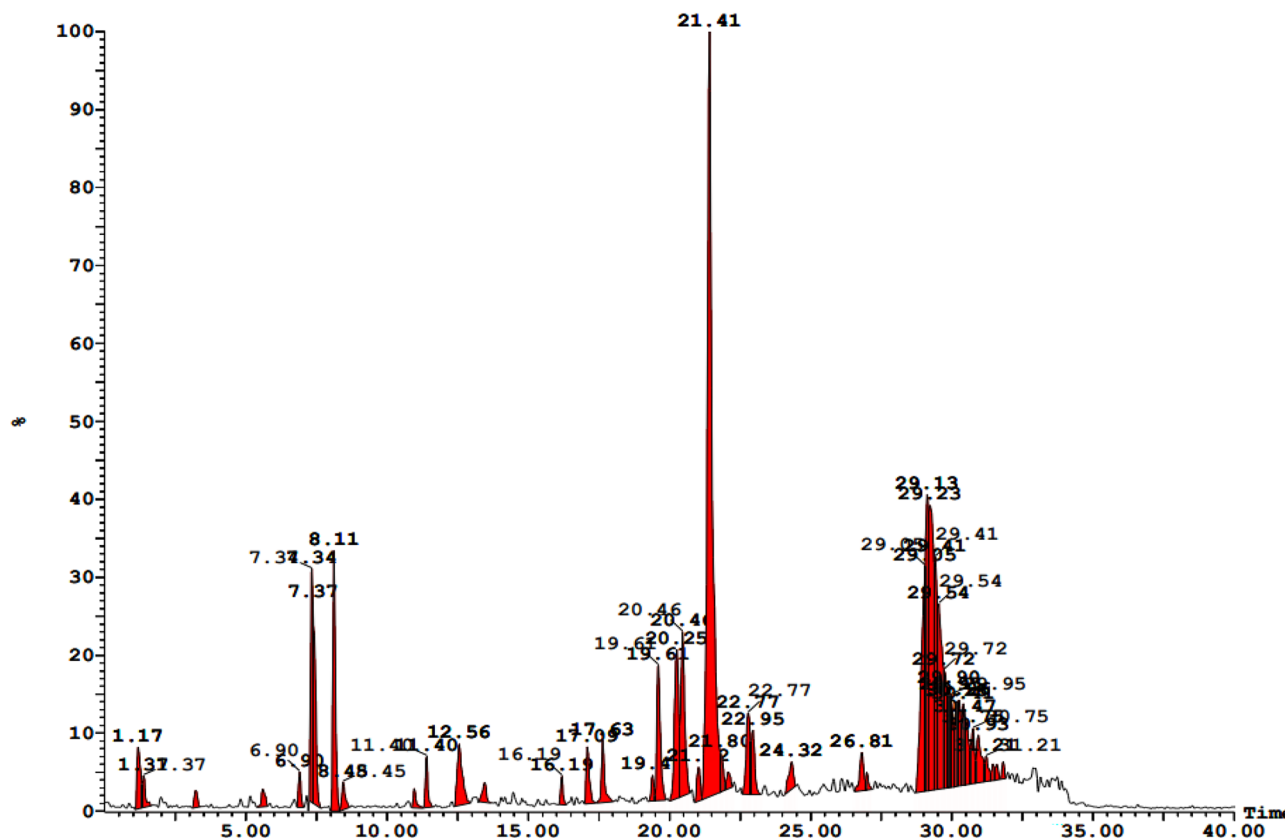


Fig. 1 Representative chromatogram of LC-MS analysis of HAENC suggesting the presence of certain phytochemicals in plant

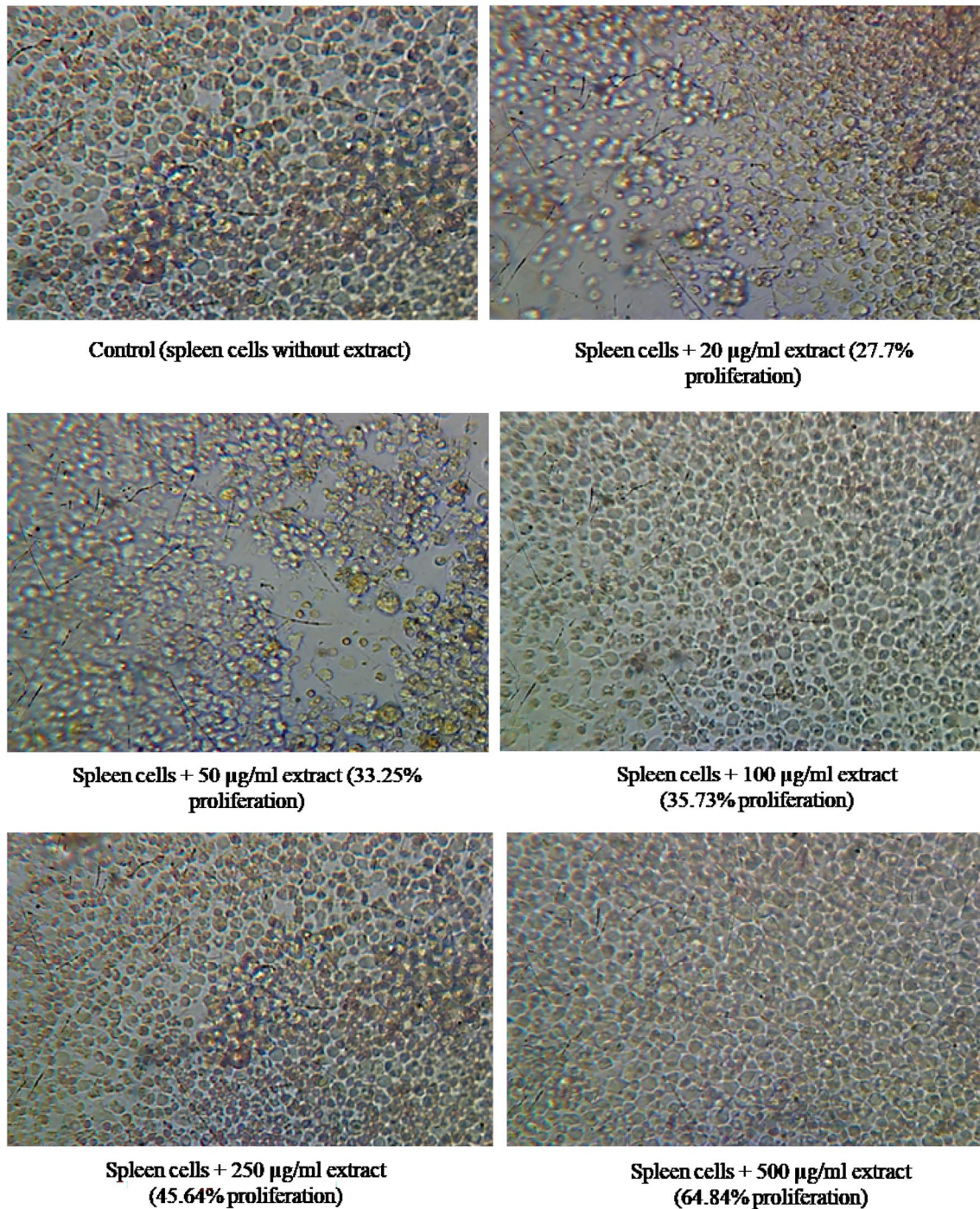


Fig. 2 Determination of cell-mediated immune response (by splenocyte proliferation) in the presence of increasing concentration of HAENC at 40× magnification

Determination of humoral immune response against *S. Typhimurium* “O” antigen /Determination of antibody titer by Indirect ELISA

Mean serum antibody titer in Wistar albino rat treated with doses 125 mg/kg, 250 mg/kg, and 500 mg/kg body weight

was found to be $2133.33^{ab} \pm 269.84$, $4266.66^b \pm 539.69$, and $8106.66^c \pm 1389.12$, respectively, when compared with the control group ($1066.66^a \pm 134.92$) as shown in Fig. 4.

Fig. 3 Determination of upregulation of anti-inflammatory (IL-10) and downregulation of pro-inflammatory (TNF- α) cytokines by Real-time PCR in the presence of HAENC

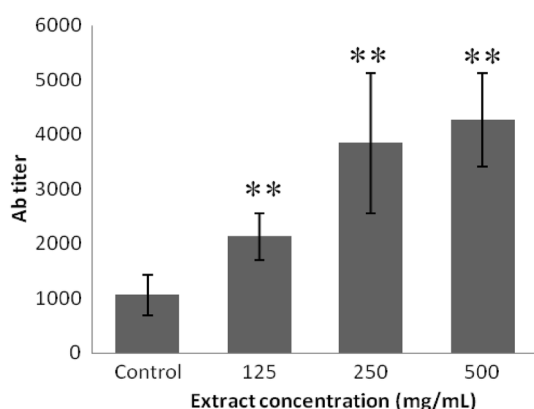
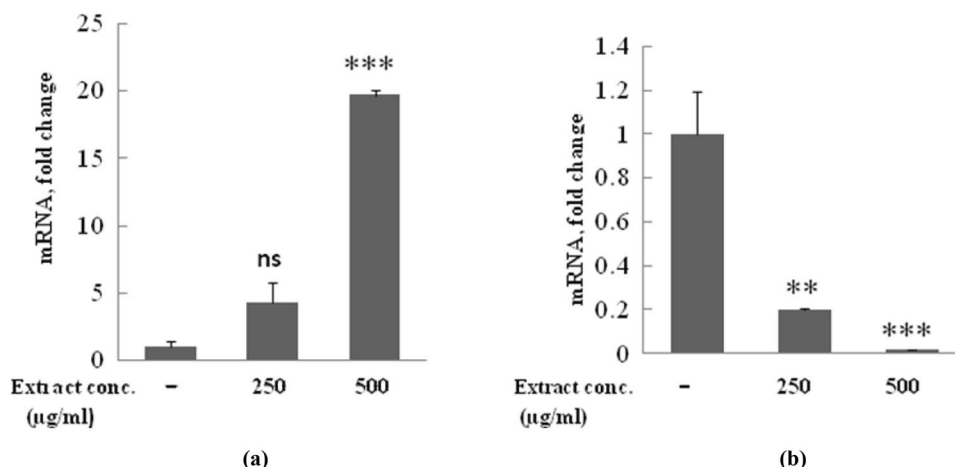


Fig. 4 Humoral immune response in Wistar albino rats using *S. Typhimurium* as "O" antigen in unfed and *N. cadamba* fruit extract-fed groups

Discussion

Recently, the concept of immunomodulation has found its position in the mainstream along with the benefits of medicinal plants as mentioned in Ayurveda by 'Charak' and 'Sushruta Samhita' due to their immunomodulatory effects (Gulati et al. 2002). Ayurveda focuses on the body's natural resistance against certain health disasters (Devasagayam and Sainis 2002). Present work was carried out to validate the immunomodulatory potential of aqueous extract of *N. cadamba* fruits.

Representative LC–MS chromatograms are shown in Fig. 1 and Table 1 and their m/z , wavelengths of maximum absorbance (λ_{max}), and retention time (RT) in Table 4.7. Comparing mass spectra and λ_{max} values from published works served as the basis for peak identification. LC–MS study of aqueous fruit extract of *N. cadamba* showed major peaks at RT 6.9, 7.37, 15.56, 17.34, 22.28, 23.74, and

24.32. Chlorogenic acid ($m/z = 354$), Octadecanoic acid ethyl ether ($m/z = 312$), and 7-hydroxy-5,2'-4'-trimethoxyflavonon ($m/z = 328$) are known for their anti-inflammatory activity. However, the current study did not confirm the presence of a particular compound and its biological action.

Current work suggests a significant increase ($p < 0.01$) in vitro cell-mediated immune potential at different doses (20, 50, 100, 250 & 500 $\mu\text{g/ml}$) of HAENC fruit concerning Wistar albino rat's splenocyte proliferation. Maximum splenocyte proliferation (64.84%) was found at 500 $\mu\text{g/ml}$ of extract of *Neolamarckia cadamba* fruit. Based on earlier (Khandelwal et al. 2015) and current studies on *N. cadamba*, it may be concluded that fruit extract might have enhanced B and T lymphocytes, which are the major elements for humoral and cell-mediated responses.

The present study depicts the enhanced humoral immune response of Wistar albino rats of treated groups with different concentrations of *N. cadamba* fruits extract (125 mg/Kg, 250 mg/Kg, and 500 mg/Kg) in comparison to the control (Fig. 4). Proposed work suggests that in presence of HAENC fruits serum antibody titer was found to increase significantly ($p < 0.01$) and was found dose-dependent, which is in confirmation with similar findings on *Kadamb*, *Brahmi* & *Guduchi* (Khandelwal et al. 2018; Husain et al. 2017). Increased antibody titer with *N. cadamba* fruit extract indicates B lymphocyte stimulation and proliferation since B lymphocyte has a major role in antibody production (Hoffman et al. 2016).

In Vitro, the effect of *Neolamarckia cadamba* fruit extract on induction of TNF- α and IL-10 was evaluated by sandwich ELISA using splenocyte culture supernatant. The study revealed that different doses (50, 100, 250 & 500 $\mu\text{g/ml}$) of fruit extract caused a significant decrease ($p < 0.01$) in the induction of TNF- α concerning the control (Table 2). A dose-dependent decrease in TNF- α production in the supernatant of Con-A stimulated splenocyte culture was observed.

Table 1 List of compounds suggested in LC–MS analysis from fruits of *N. cadamba*

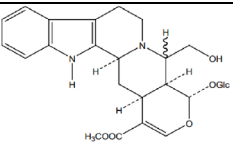
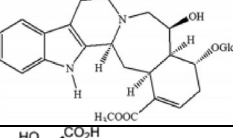
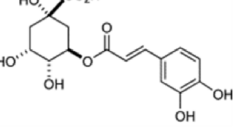
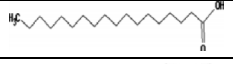
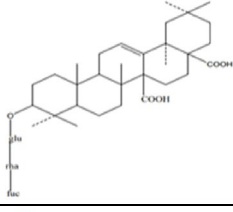
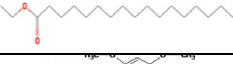
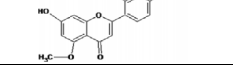
S.No.	Name of compound	Chemical formula	Structure of the compound	m/z ratio	Adduct	Function	References
1	Isohydrocadmbine	C ₃₇ H ₄₄ N ₂ O ₁₅		546	[M+H] ⁺	Antileishmanial, Antidiabetic, Antiplasmodial	Pandey and Negi, 2016
2	Dihydrocadmbine	C ₂₇ H ₃₄ N ₂ O ₁₀		546	[M+H] ⁺	Antileishmanial, Antidiabetic, Antiplasmodial	Pandey and Negi, 2016
3	Chlorogenic acid	C ₁₆ H ₁₈ O ₉		354	[M+H] ⁻	Antioxidant, Antiinflammatory, Antiobesity, Antidiabetic, Antimutagenic, Endothelial protective, Antilipidemic, Antifibrosis, Hypotensive	Kapil et al., 1995; Li and Chang, 2005; Thom, 2007; Meng et al., 2013; Shi et al., 2009
4	Stearic acid	C ₁₈ H ₃₆ O ₂		284	[M+H] ⁻	Cardioprotective	Ding et al., 2006
5	Cadambagenic acid	C ₃₀ H ₄₆ O ₅		486	[M+H] ⁻		Mondal et al., 2020; Sahu et al., 1974
6	Octadecanoic acid ethyl ether	C ₂₀ H ₄₀ O ₂		312	[M+H] ⁻	Anti-inflammatory	Shashikant et al., 2022
7	7-hydroxy-5,2'-4'-trimethoxyflavonon	C ₁₈ H ₁₆ O ₆		328	[M+H] ⁺		Mondal et al., 2020

Table 2 Upregulation/downregulation of IL-10 and TNF- α in the presence of HAENC

S. no.	Extract Concentration	Concentration of TNF- α (pg/mL)	% inhibition	Concentration of IL-10 (pg/mL)	% stimulation
1	Control (without extract)	2776.33 ^e $\pm$ 46	–	3553.33 ^a $\pm$ 69	–
2	50 μ g/mL	2548.16 ^d $\pm$ 43	8.21%	3916.66 ^b $\pm$ 77	10.22%
3	100 μ g/mL	1794.16 ^c $\pm$ 52	35.37%	4161.66 ^{bc} $\pm$ 118	17.11%
4	250 μ g/mL	1334.16 ^b $\pm$ 37	51.94%	4336.66 ^c $\pm$ 95	22.04%
5	500 μ g/mL	818.33 ^a $\pm$ 48	70.52%	4901.66 ^d $\pm$ 95	37.94%

TNF- α has been linked to several autoimmune and inflammatory conditions, including uveitis, multiple sclerosis, Crohn's disease, and rheumatoid arthritis (Dan et al. 2021). Perseverance of TNF- α will cause harmful changes in the metabolism of lipids and glucose and cause chronic inflammatory conditions (Popa et al. 2007). The present study concludes that *N. cadamba* fruit extract downregulates TNF- α level, which provides scientific evidence to use against diseases arising due to chronic inflammation. Downregulation of TNF- α in splenocyte culture with different doses of *N.*

cadamba fruit extract was further confirmed by qRT-PCR analysis. Fold change in the expression of target gene TNF- α was 0.197 and 0.02 at 250, and 500 μ g/mL fruit extract concentrations respectively in comparison to control (untreated) TNF- α , suggesting the decreased mRNA expression of TNF- α cytokine gene at 250 and 500 μ g/mL (Fig. 3).

N. cadamba fruit extract-treated splenocytes showed a significant increase ($p < 0.01$) in the induction of IL-10 in comparison to the control (Table 2). IL-10 is a powerful anti-inflammatory cytokine, essential in preventing

inflammatory and autoimmune pathologies (Bazzoni et al. 2010). IL-10 causes proliferation and differentiation of B cells which leads to enhancement in the level of IgM, IgG, and IgA antibodies (Agrawal et al. 2009). Present findings suggested that HAENC fruit enhances humoral immune response by augmenting the level of IL-10 cytokine. Upregulation of IL-10 in the presence of different doses of *N. cadamba* fruit extract in Con-A induced splenocytes culture was further confirmed at gene level by real-time PCR. Fold change in expression of target gene IL-10 was 3.58 and 20.11 at 250 and 500 µg/mL fruit extract concentrations respectively in comparison to the control (untreated) IL-10 gene (Fig. 3). An earlier study (Platzer et al. 1995) suggested that Upregulation of IL-10 was found to decrease the level of TNF- α . Our results also complement these findings and the upregulation of IL-10 and downregulation of TNF- α confirms the anti-inflammatory effects of *N. cadamba* fruits at the protein and gene level.

The findings of the present study demonstrate that fruit extract from *N. cadamba* significantly increased splenocyte proliferation and regulated the networking of pro-inflammatory cytokines (TNF- α) and anti-inflammatory cytokines (IL-10) to maintain the body's homeostasis, suggesting immunostimulatory potential. Furthermore, the dynamic regulation of cytokine induction and expression used in this study supports the use of *N. cadamba* fruit in treating a variety of disorders.

Conclusion

The above study confirms the potential benefit of *N. cadamba* fruit concerning its immunomodulatory mechanism indicated by cytokine expression and increased antibody titer. Increased level of IL-10, decreased level of TNF- α at gene, as well as protein level suggest its anti-inflammatory potential in animals. This study applies scientific techniques including sandwich ELISA, qRT-PCR, and in vivo experimentation to confirm the immunomodulatory potential of *N. cadamba* fruits. The current study is restricted to confirming the presence of a particular compound and its chemical configuration. There is a need to find out particular compounds present in the extract that can inhibit TNF- α and increase IL-10. Furthermore, experiments and clinical trials following precise and standard methodology are mandatory to validate the immune-modulatory mechanism to analyze the medicinal potential of plants.

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Data availability The data supporting the findings of the work are presented in various tables and figures and are available within the article.

Declarations

Conflict of interest The author(s) declare that they have no potential conflicts of interest concerning the research, authorship, and/or publication of this article.

Ethical approval Animals were obtained from the animal house of ICAR IVRI, Izatnagar Bareilly (India) and were kept with standard management support in the animal house of GLA University with IAEC approval vide GLAIPR/IAEC/2021–09/06.

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